Intact

A 30 s video over a 20–100% network, accepted only when every byte matches.

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The backend checks the whole-file SHA-256 before it touches the video, then decodes every frame, finds missing, duplicated and damaged frames at 30 FPS, and says whether the evidence points to transfer, capture or encoding.

50/50 uploads finished intact under 10 bad-network scenarios, SHA-256 matched every time	**0.933** recall: share of 223 known faults found in 120 held-out videos	**0.995** precision: 1 false alarm in 194 detections	**82** automated tests, one command: `pytest -q`
75 faults injected mid-transfer	**0** damaged or partial files accepted (16 tests)	**0** false alarms on 25 clean videos	**5** real browser recordings, incl. iPhone

Three rules the system never breaks

Accept only intact files
COMPLETE only when the server's SHA-256 of the whole file equals the client's

Analyse only accepted files
the analysis can open a file only through a check that refuses anything else

Never guess the cause
no evidence → "cannot determine", never "no drops"

What is in the repository

Code
Python uploader, browser recording page, FastAPI backend, analysis pipeline

Evidence
raw logs of all 50 network runs, 120 scored videos, 5 browser runs

Reports
sample integrity reports (HTML + JSON), design notes, limitations, demo script

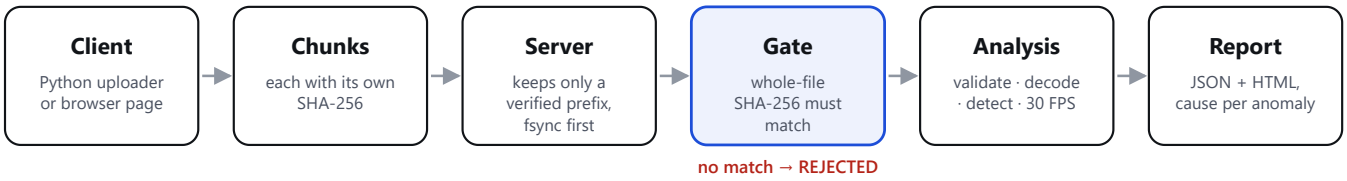
Evidence the brief asks for, and where it is

✓	**Stable connection and variable signal (20%–100%)**	15 stable + 5 fluctuating runs, all verified	p. 3
✓	**Temporary interruption, timeout and reconnection**	outage, resets, hangs, client and server kills: 25 runs, all verified	p. 3
✓	**Connection switching or fluctuating speeds**	5 switch + 5 fluctuating runs (emulated; see limits)	p. 3, 6
✓	**Altered, truncated or invalid video**	120 held-out videos scored, 3 samples end to end	p. 4, 6
✓	**Architecture, README and run instructions**	pipeline, states, commands; README in the repository	p. 2, 6
✓	**Sample integrity report and limitations**	HTML + JSON reports; limits stated plainly	p. 6

repository: link in the submission email. Every number in this document is read from a file there.

One recovery rule: ask the server, continue from there

The upload is cut into chunks. The server keeps only a contiguous, verified prefix of the file, so after any failure the right place to continue is simply the number of bytes it holds.



An upload, step by step

1. **Open a session.** The client declares the file size and its SHA-256.
2. **Send chunks.** Each goes to a byte offset with its own SHA-256. A damaged chunk is refused; an identical re-sent chunk changes nothing.
3. **Store safely.** The server writes the chunk, calls fsync, and only then records the new offset in SQLite, so the recorded offset is never ahead of the disk.
4. **Recover.** Lost network, timeout, client crash or server crash: the client asks for the offset and continues.
5. **Finalize.** The server re-reads and re-hashes the whole file. Only a match gives COMPLETE; analysis starts automatically.

Behaviour on a bad link

- chunk size **64 KB – 8 MB**, doubles after a success, halves after a failure
- retry: exponential backoff, full jitter, cap **30 s**
- no progress for **20 s** → drop the connection, retry
- up to **15** failures in a row before giving up

Server states

UPLOADING	chunks arriving
STALLED	no completed chunk for 30 s: a timer, not a verdict
FAILED	no chunk for 24 h: session closed
COMPLETE	whole-file SHA-256 matches
REJECTED	assembled file does not match: never analysed

```
COMPLETE ⇔ SHA-256(file on server disk) = SHA-256(file the client declared)
```

A real recovery, from the raw log (interruption scenario, run 1)

1 — Send
chunk at byte `1,310,720` (512 KB)

2 — Cut
the server stores it; the connection drops before the reply

3 — Ask
client retries: "how many bytes?" → `1,835,008`

4 — Continue
from `1,835,008`. Nothing sent twice

50 of 50 uploads arrived byte-perfect

Run on a separate Linux laptop: netem shapes a private virtual link in both directions, Toxiproxy cuts, resets and hangs connections. Each fault fires when the client's own log shows a set fraction of the file sent, so it always lands mid-transfer. Test video 3,756,860 bytes, chunks up to 512 KB.

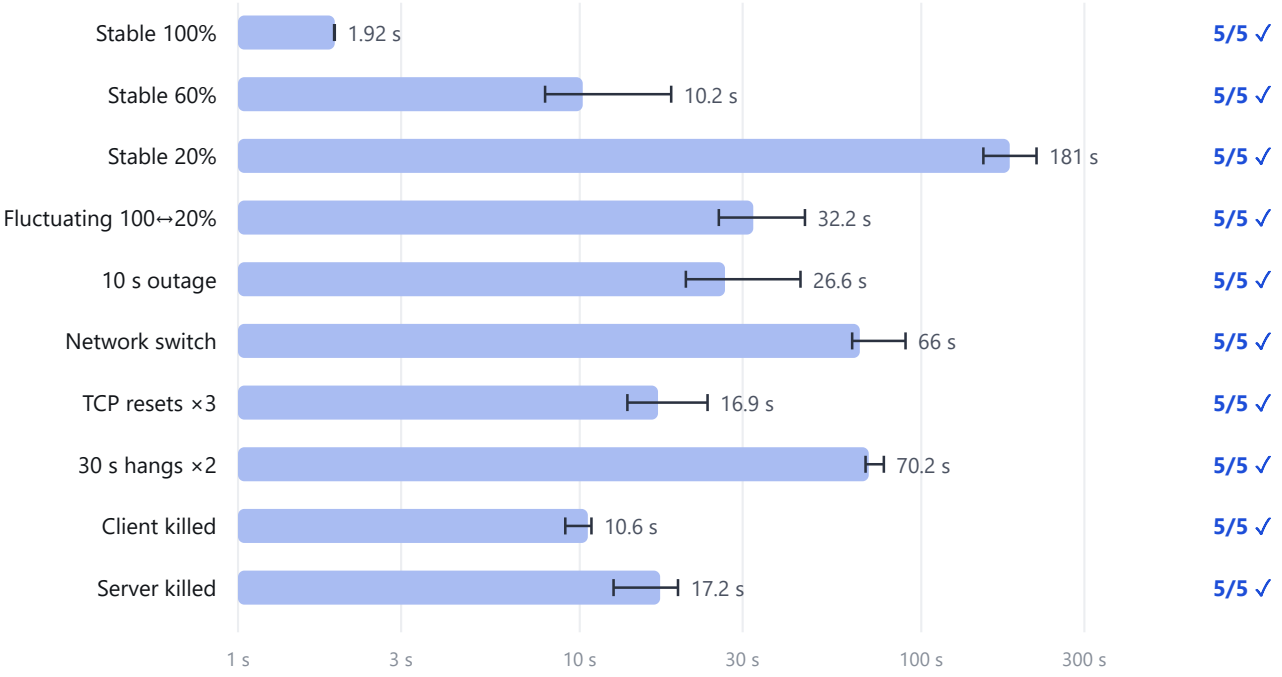
100%
20 Mbit/s · 20 ms RTT · 0% loss

60%
5 Mbit/s · 80 ms RTT · 1% loss

20%
500 kbit/s · 400 ms RTT · 5% loss

Measured before the runs: ping 22.2 ms at 100%, 401.7 ms at 20%.

Upload time per scenario: median of 5 runs, whiskers min–max (log scale)



13
times a chunk arrived but its acknowledgement was lost. The client asked, skipped ahead, and never sent that chunk twice.

≤ 512 KB
re-sent after the server was killed (SIGKILL) mid-upload and restarted, over 5 runs.

140–200 kbit/s
is what the 20% link really carries (5% loss, 400 ms). Every upload still completed, in 152–217 s.

Checked separately from the harness: in every run the acknowledged chunks plus the server's resume offsets cover exactly bytes 0 to 3,756,860. 151 retries in total, 0 harness errors.

What the fault scenarios do

Fluctuating 100↔20%
100 → 60 → 20 → 60 → 20 → 100% at 15, 30, 50, 65, 80% of the file

10 s outage
60% link; at 30% every connection is cut and refused for 10 s

Network switch
100% link drops for 3 s at 40% and returns as a 20% link (client address unchanged)

TCP resets ×3
60% link; every connection reset (TCP RST) for 2 s at 25, 50, 75%

30 s hangs ×2
60% link; data silently dropped for 30 s at 30% and 60%, longer than the 20 s stall timeout

Client killed
client process SIGKILLed at 40%, restarted 2 s later

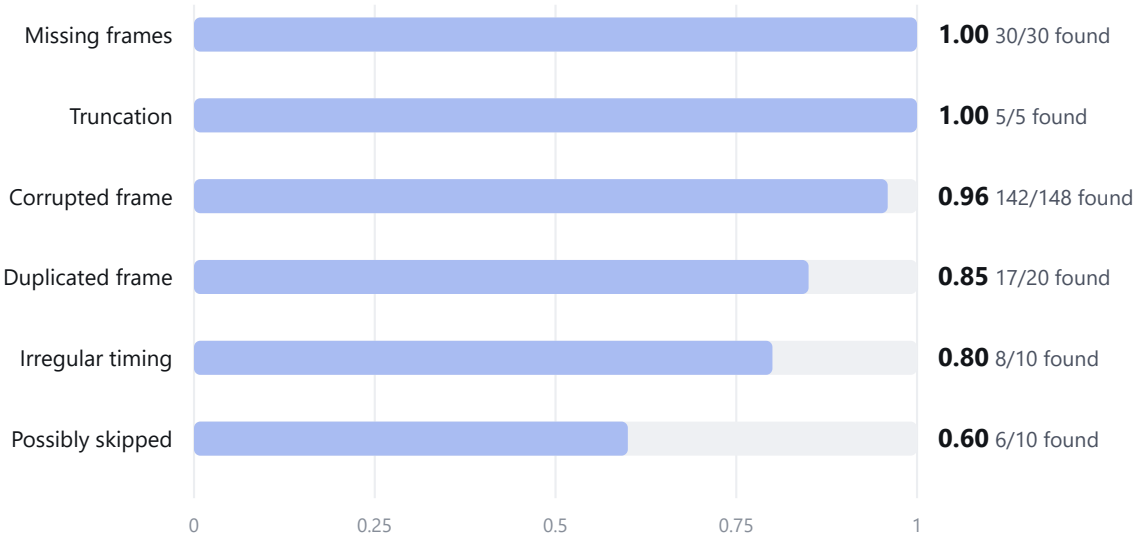
Server killed
server process SIGKILLed at 40%, restarted 3 s later

Found 208 of 223 known faults; 1 false alarm

24 kinds of faulty 30 s video with known ground truth. Thresholds were tuned on seed 1, the code frozen, then seeds 2–6 (120 new videos) scored once. Every miss is listed in the repository.

0.933	0.995	100%	0
recall	precision	of videos judged valid / invalid correctly	false alarms on 25 clean control videos

Recall by anomaly type



How a file is checked

1. **Validate:** ffprobe reads the container and streams.
2. **Decode everything:** PyAV decodes every frame and records every decoder message; the ffmpeg CLI decodes again as an independent check.
3. **Detect on real timestamps:** gaps, repeats, jitter, damage, truncation; each with time, frame indices, evidence, confidence.
4. **Then 30 FPS:** map each slot to a source frame; repeats made by that conversion are labelled artefacts, not defects.

What it misses, and why

6	byte flips that decode without any error and leave no visible trace (damage in transit is caught by the hash)
4	frames dropped with re-generated timestamps: no gap left, only a motion jump
3	duplicates in near-still scenes, where a repeat looks like no motion
2	timing jitter just under the 25% threshold

What one finding looks like (from the sample report)

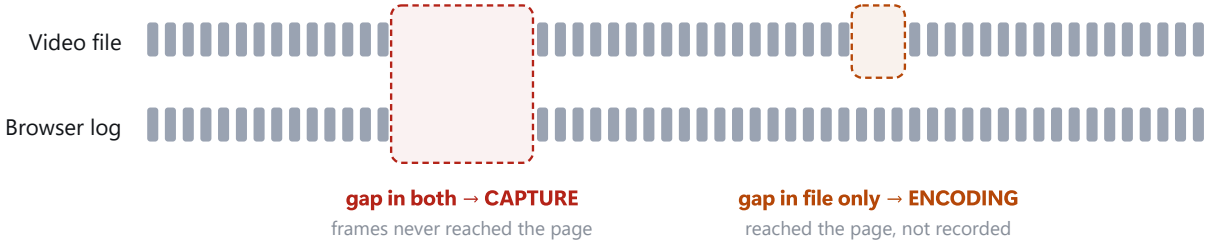
`missing_frames` `11.833 – 12.067 s` `frames 355–356` `confidence: high`

```
“interval 233.33 ms = 7.00x nominal 33.33 ms; ~6 frame slot(s) empty between 11.8333s and 12.0667s with no packet in the container”
```

likely cause **undetermined** · candidates capture, encoding · ruled out **transfer** (the hash matched)

A matching hash rules out the network. The frame log tells capture from encoding.

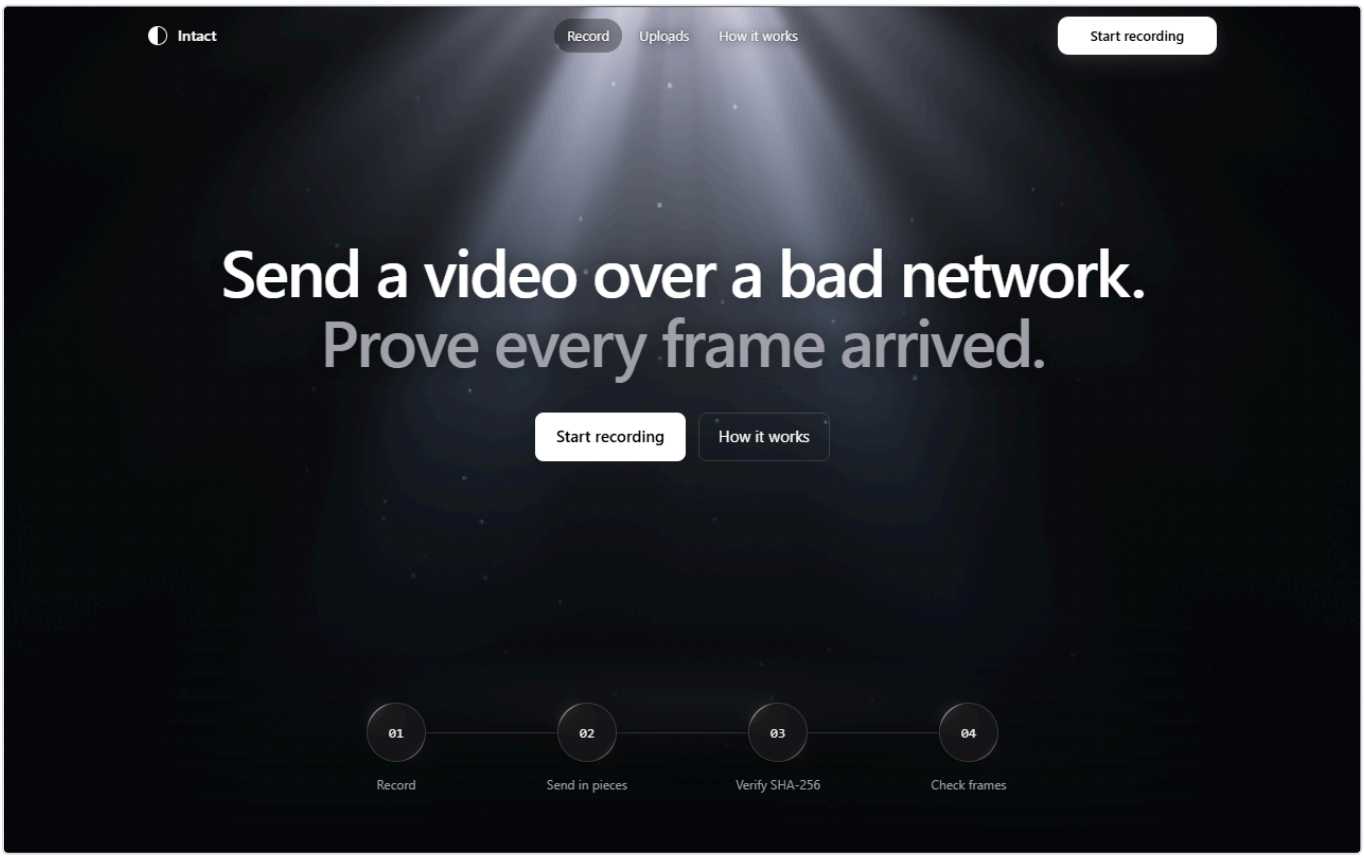
The browser page logs every frame that reaches it (`requestVideoFrameCallback`) and uploads the log with its own hash. The log is aligned to the file by a time search, and used only when the frame pattern or a shared gap pins the offset down; otherwise the report says the alignment is ambiguous.



Found in real recordings

RUN	EVIDENCE	VERDICT
Webcam, desktop Chrome	**3.3 s** gap (102 frames) at 0.30–3.60 s, in the file **and** the log	capture
Webcam, desktop Chrome	**2** frames shown in the page but missing from the file	encoding
Injected 0.5 s source stall	file gap 10.046–10.575 s; log gap **529.5 ms** with no frames inside	capture
iPhone, Safari	**899/899** frames aligned, **0** delivery gaps; recorder started in **3 ms** (Chrome: 1.5–3.8 s)	no drops

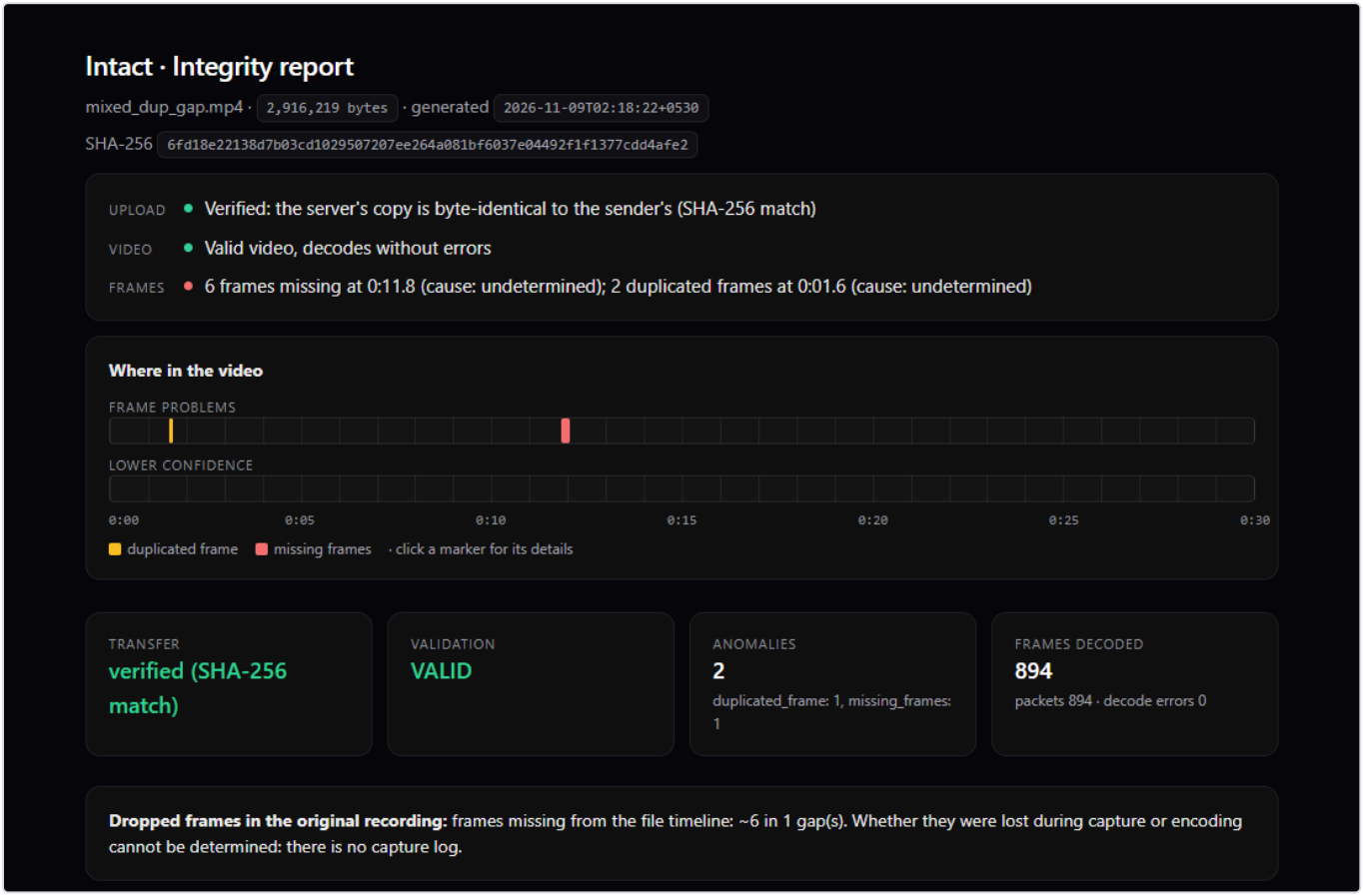
The recording page



Record 30 s, watch it upload in verified pieces, read the verdict. Animations stop while recording and uploading so they never compete with the frame log.

Both planted faults found, at the right frames

A 30 s video with frame 47 repeated twice and frames 356–361 removed, uploaded through the system. No capture log exists for this file, so the cause stays *undetermined* rather than guessed.



docs/samples/mixed_dup_gap.html. Also in the samples: a file cut to about half (VALID_WITH_ERRORS, truncation located at 15.33 s) and a text file named .mp4 (INVALID).

Limits, stated plainly

- **A perfect transfer does not prove a clean recording.** Frames dropped before encoding with new timestamps leave no gap.
- **The frame log is written by the client:** its integrity is checked, its authenticity cannot be. Resolution about 17 ms.
- **Detector thresholds are fixed;** near-still scenes and silent byte flips are its known misses.
- **Network conditions were emulated.** I did this alone during end-semester exams, which left no time for a real mobile-network or real connection-switch test. Next test: server on a second computer, both on one router, the laptop switching between its 2.4 and 5 GHz networks mid-upload so its address really changes. Pass: COMPLETE, hashes match, at most one chunk re-sent.
- Not tested: Firefox, the browser client under netem, resuming after a page reload. No authentication.

Run it

```
$ pip install -r requirements.txt   # Python 3.12/3.13, FFmpeg on PATH
$ python -m pytest -q   # 82 tests
$ uvicorn server.app:app --port 8000   # then open http://127.0.0.1:8000/capture
$ python -m client.uploader video.mp4 --server http://127.0.0.1:8000
```

Two WebGL effects on the recording page are ports of React Bits components (MIT + Commons Clause; licence file in server/static/).